

Assignment-1: Key Deep Learning Concepts

Marks: 3x10 = 30

Instructions:

- This assignment covers core concepts that are part of your **syllabus** and will appear in the **exam**.
- Your task is to **understand** each concept and write a short explanation (2–3 lines) in your own words.
- For each concept, include:
 1. The main idea in simple language.
 2. The key mathematical function or equation.
 3. A small example calculation (with numbers where possible).
- Do not copy-paste from ChatGPT or other sources. If you do, you will lose the opportunity to truly learn these concepts and risk poor performance in the exam.

Topics to Cover:

1. **Batch Normalization**
2. **Vanishing Gradient Problem**
3. **Exploding Gradients Problem**
4. **Leaky ReLU**
5. **Dying ReLU Problem**
6. **Exponential Linear Unit (ELU)**
7. **Dropout Regularization**
8. **1×1 Convolutions (Pointwise Convolutions)**
9. **Residual Connections and Identity Mapping**
10. **Depthwise Separable Convolutions**

Example of What Your Answer Should Look Like:

- **Concept:** Sigmoid Function
- **Explanation:** Sigmoid maps any real number to a value between 0 and 1. Useful for probabilities.
- **Function:** $\sigma(x) = 1 / (1 + e^{(-x)})$
- **Example:** If $x = 0$, $\sigma(0) = 0.5$